When do GPT-2's attention heads rest?

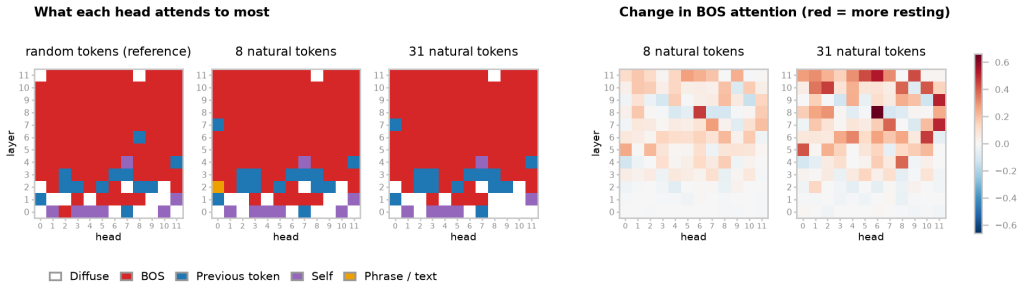
GPT-2 small · BOS + 32 tokens · common filler shown

Idea: late heads rest on BOS (a no-op) when the context has nothing for the current token to use, and stop when the current token completes it.
Examples below are real rows: BOS start token gray random filler blue inserted words orange final token, where attention is measured.

① Experiment 15 · more coherent text → more resting

1:00–7:00

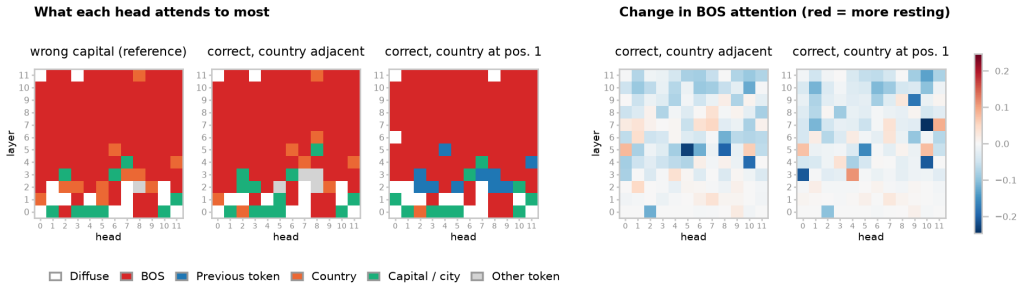
Tested	1 to 31 words of natural text before a random final token.
Hypothesis	Resting levels off after about 3 words.
Confound	Fewer random tokens: matched with scrambled words and look-alike random words.
Result	**Wrong:** resting keeps rising, +0.13 / +0.16 at 31 words, even without the last word pair.
Example	8 words: BOS ir ations 01 … 19 random … ---- he knew that tomorrow the weather would be count 31 words: BOS sand as the sun went down … 21 words … the weather would be count
Graph	Most heads already rest on BOS; the change maps turn redder as the text gets longer.



② Experiments 9–10 · completion releases it

7:00–13:00

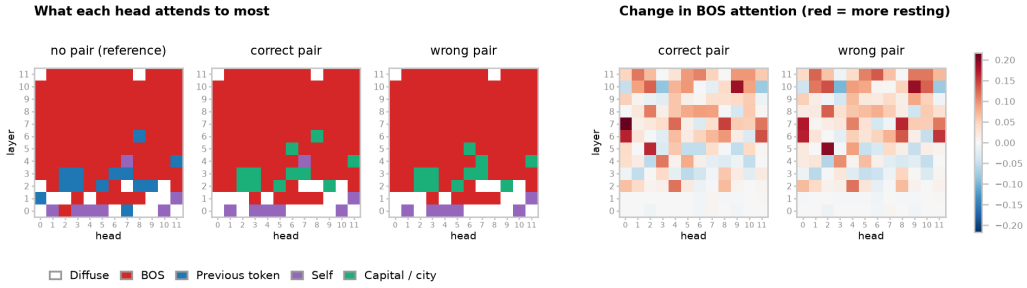
Tested	Capital as the final token, with its country 1 to 31 positions earlier.
Hypothesis	A correct capital pulls late heads off BOS.
Confound	Final-token identity: each capital used once correctly and ten times wrongly.
Result	**Right:** -0.018 / -0.034 in 11/11 countries, from every position; placebo 0 of 496.
Example	correct: BOS she sc sa … 27 random … France Paris wrong: BOS she … 29 random … Japan Paris far: BOS France sc sa … 27 random … dis Paris
Graph	Late heads turn blue (less resting) for correct capitals, whether the country is adjacent or far.



③ Experiment 8 · truth alone does not

13:00–19:00

Tested	True vs false country–capital pair before a random final token.
Hypothesis	A true pair raises resting more than a false one.
Confound	Token identity: full country × capital grid; GPT-2 knows all 11 capitals.
Result	**Wrong:** true ≈ false (-0.001 / -0.004); any real pair raises resting (+0.013 / +0.05).
Example	true: BOS she sc sa … 26 random … France Paris ious false: BOS she sc sa … 26 random … Japan Paris ious
Graph	Correct and wrong pairs give the same red change vs no pair.



Proposed Experiment 16 · is resting actually a no-op?

19:00–29:00

Tested	Remove each late head where it rests vs where completion releases it.
Hypothesis	Resting heads write almost nothing, so removing them barely changes predictions.
Confound	Token changes: reuse grids where resting moves but the tokens stay fixed.
Why	①–③ measure attention only; this tests whether resting really does nothing.
Example	The same rows as ① (31 words vs random tokens) and ② (correct vs wrong capital).
Graph	Planned: cost of removing each head against its BOS attention, resting vs released.

Trust: pre-registered predictions (19 of 37 held) · TransformerLens = Hugging Face to 5×10^{-5} · placebos quiet · copying heads detected as a positive control. **Limits:** GPT-2 small only · mostly random-token contexts.